

Diabetes Prediction Using Machine Learning and Hybrid Deep Learning Ensemble Technique

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Abstract— *Diabetes prediction is an essential task of healthcare which enables early diagnosis and treatment. In this study, we present a comprehensive analysis utilizing machine learning and ensemble deep learning techniques for diabetes prediction, leveraging two distinct datasets: such as the PIMA diabetes dataset which contains 768 records with 9 attributes, and another diabetes dataset which has exactly 2000 records with 9 attributes. The approach that we have adopted for cleaning up data has a lot of prerequisites, aimed at achieving accuracy. We applied a preprocessing technique for removing duplicates and replacing zero values with the mean or median. In addition to that, we applied the scaling process known as standardization on our given data, it is a method that ensures the average of all the values is 0 and the standard deviation is 1. To process our input finding, we implemented a set of machine learning algorithms covering K Nearest Neighbors, Naive Bayes, Support Vector Machine, Decision Tree, Random Forest, and Logistic Regression along with ensemble deep learning techniques of TabNet, XGBoost, and Multilayer Perceptron which shows the best possible outcomes. The proposed hybrid deep learning ensemble approach presents the best results for predicting diabetes. In our research hybrid deep learning ensemble technique has accuracy of 96%, this method shows much higher efficiency over common machine learning algorithms.*

Keywords— *diabetes prediction, deep learning, machine learning, ensemble learning, hybrid model, healthcare predictive analytics, medical data analysis.*

I. INTRODUCTION

Diabetes Mellitus, a chronic metabolism condition defined by hyperglycemia or elevated blood sugar level, poses a great public health problem globally. Identifying and controlling diabetes in the initial stage is essential as it can help in delaying onsets of fatalities and major health problems. This study seeks to review meticulously the applications of both ML and DL models for diabetes prediction to assist practitioners in improving early diagnosis and the approaches towards handling the complications as a result of diabetes. A study discussed the automation of Diabetes detection systems using machine learning and deep learning techniques to come up with more efficient and intelligent healthcare systems [1]. According to the International Diabetes Federation, there were 463 million people with diabetes in 2019, it will reach 700 million by 2045 [2]. Machine learning (ML) and deep learning (DL) ensemble techniques have been deemed innovative methods that can increase the precision of diabetes prediction. This study aims to investigate the effectiveness of ML algorithms and ensemble DL models in predicting diabetes onset,

utilizing two comprehensive datasets: the Pima Indian Diabetes dataset consisting of 768 records and the second dataset consisting of 2000 records and 9 attributes. A recent study applied machine learning and deep learning techniques including stacking ensembles for diabetes classification using the PIMA Indian Diabetes dataset and found better accuracy and performance for the predictions [3]. Several machine learning algorithms were applied to the Pima Indian Diabetes dataset such as Logistic Regression and these algorithms achieved very good accuracy depicting the capability of these techniques in the prediction of diabetes [4]. Data preprocessing involves the strict examination of data quality by removing duplicates for a fair comparison, and imputation of missing values by mean and median procedures to ensure the preservation of data integrity. In addition, we were able to perform a scaling process known as standardization on our data set. We also use different machine learning approaches such as K-Nearest Neighbors, Naive Bayes, Support Vector Machine, Decision Tree, Random Forest, and Logistic Regression that are best suited to get better results. Additionally, ensemble DL techniques such as TabNet, XGBoost, and Multilayer Perceptron are applied so this should help to utilize the collective output of all the models.

II. RELATED WORKS

Over the past few years, machine learning and deep learning methods applied to diabetes prediction have been a subject of curiosity by the research community. Numerous studies were conducted regarding the development of different approaches for improving diabetes forecasting performance by using several data sources and algorithms. In V. Chang, J. Bailey, Q [5] research, the Pima Indian diabetes dataset is employed to assess the machine learning approaches to predict diabetes as practiced in other similar studies. It is based on the similar methods to improve the prediction accuracy using different approaches to machine learning.

Gangani Dharmarathne [6] presented an easy-to-understand interface to diagnose diabetes using machine learning models namely Decision Tree, KNN, SVC, and XGB. Their approach employed Shapley Additive Explanations for model interpretability where in XGB model the result was quite promising. They incorporated SHAP in an interface to provide clear diagnosis and improve users' awareness. I. Tasin [7] built a diabetes prediction system utilizing different machine learning techniques and a private

dataset and employed other methods like SMOTE and SHAP to increase the efficiency of the model and make it more intelligible. However, the present work employs a machine learning and deep learning approach, namely TabNet and XGBoost with an aim of enhancing performance through similar preprocessing steps. Ritika Bateja [8] proposed a diabetes prediction system using SVM enhanced with Particle Swarm Optimization and Collaborative Filtering for drug recommendations. They utilized data cleaning, null value removal, and feature selection for preprocessing. Inspired by their work, our research adopts the SVM algorithm to enhance our diabetes prediction model. J. Abdollahi [9] carried out feature selection by PSO in diabetes prediction and further studied three diverse domains from medical area with different machine learning approaches as a result. Their results showed Decision Tree, Random Forest and Naïve Bayes to be most successful in accuracy as well as error rate. Cindy Nabila Noviyanti [10] worked on enhancing the accuracy for the early detection of diabetes with help from Random Forest algorithm. They achieved an accuracy of 87% with the Pima Indian Diabetes dataset, emphasizing the effectiveness of Random Forest in improving prediction performance. We use this in diabetes research to optimize early detection methods including Random Forest. Through the performance of experiments and comparative analysis, the researcher determined the different criteria for various algorithms that are commonly used in measuring the level of risk of diabetes.

In these studies traditional machine learning models such as Support Vector Machines and k-nearest Neighbors for diabetes prediction. Their models often struggle with large complex datasets and may not apply to diverse patient groups. Additionally, they usually focus only on accuracy, overlooking important metrics like precision and recall. In this study, we used a hybrid deep learning ensemble approach to overcome these issues, improving prediction performance and offering a more balanced evaluation of various metrics that enhance the model's practical use in clinical settings.

III. METHODOLOGY

A. Data Description

In this study, we utilize two different Diabetes datasets. The first one called Pima Diabetes has 768 records with 9 attributes [11], while the second dataset is 2000 records with the same number of 9 attributes [12].

Table 1. Dataset Information

Attributes	Description	Data Type
Pregnancies	No. of time pregnant	int
Glucose	Plasma glucose concentration a 2 hours in an oral glucose tolerance test	int
BloodPressure	Diastolic blood pressure (mm Hg)	int
SkinThickness	Triceps skin fold thickness (mm)	int
Insulin	2-Hour serum insulin (mu U/ml)	int

BMI	Body mass index (weight in kg/(height in m) ²)	float
DiabetesPedigree Function	Diabetes pedigree function	float
Age	Age (years)	int
Outcome	Class variable (0 or 1)	int

The datasets utilized in this study are from reputable sources and are frequently used in diabetes research as well. They were chosen because of their differences in size and attributes which helped in a thorough testing of the proposed methods. In addition, preprocessing was carried out extensively on the datasets to confirm the credibility and reliability of data for the succeeding analyses. Table 1 describes the dataset, which have the description of all the attributes and their data types.

B. Data Preprocessing

In preprocessing, we applied techniques such as removing duplicates, handling missing values, and scaling the data. These methods were chosen to preserve essential attributes and maintain data integrity. We implemented these techniques specifically to avoid dimensionality reduction and data loss to ensure that the raw data remained intact for our analysis.

- **Removing Duplicates:** The elimination of duplicates guarantees the authenticity of the data, and we do not want the study to be biased by the repeated counting of duplicate records.
- **Handling Missing Values:** Instead of zero values, we can replace them with means in 'Glucose' and 'BloodPressure', and in 'SkinThickness', 'Insulin', and 'BMI', the medians can be used. This effectively handles missing data, and it is useful for future analysis.
- **Scaling the Data:** The third and last step among the preprocessing pipeline was feature scaling which was performed with the help of StandardScaler module in such a way that the features were normalized by removing the mean as well as scaling it to unit variance. This standardizes the data and places it on a scale of 0 mean and 1 standard deviation. Standardization minimizes the risk of any feature dominating the training by bringing the features to a unified level thus improving the rate of convergence of most algorithms such as k-nearest Neighbors, Support Vector Machines, and even Neural Networks. Through numerical scaling and normalization, it addresses issues with handling features with large ranges or in different units, which can have a larger impact on the model. Standardization leads to more balanced and reliable model performance.

C. Model Building

In our model, we can see from figure 1 is actually based on a machine learning model as well as a deep learning model which are both ensembles. The machine learning algorithms applied are KNN (K Nearest Neighbors), Naive Bayes, SVM (Support Vector Machine), Decision Tree, Random Forest, and Logistic Regression.

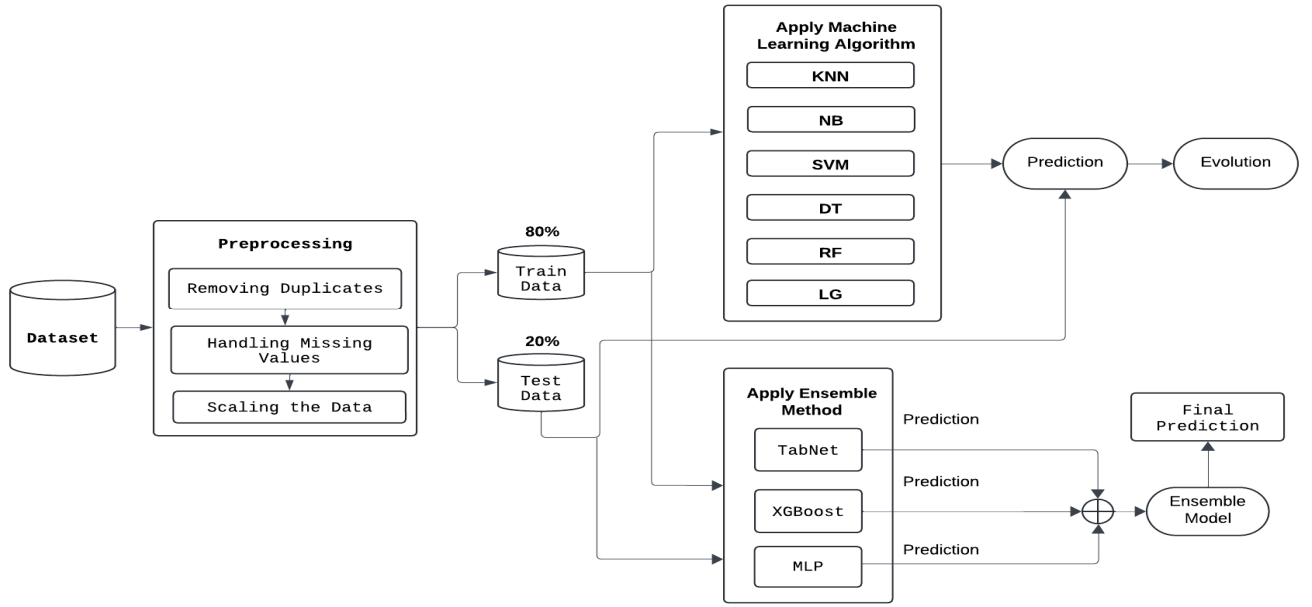


Fig. 1. Model Architecture

Besides that, we used Deep Learning algorithms in ensemble models such as Tabnet, XGBoost, and Multilayer Perceptron.

We developed a unique predictive model by testing different combinations of methods. After comparing the results with standard machine learning algorithms. The final combination of MLP, TabNet, and XGBoost gave the best results. In previous studies, for diabetes prediction, several machine learning algorithms like DT, SVM, and LR. KNN and RF have been used which is in correlation with the models used in this study [13][14][15]. Our goal is to improve the model's performance in predicting diabetes.

Before applying these algorithms, first we take the dataset and apply various preprocessing techniques. There are removing duplicates, scaling the data. These preprocessing steps have the potential to improve the quality, consistency, and functionality of datasets, ultimately improving the dependent performance and accuracy of insights obtained during analysis or modeling. After that, splitting both of the dataset into training and testing is 80% and 20% respectively. Then train the models with the training dataset and prediction comes out from the testing dataset.

1. K Nearest Neighbors

K-Nearest Neighbors is an instance-based learning technique, which makes predictions about the probability of particular class by taking into consideration the majority value of K nearest most data points. It classifies a new data point by finding the K closest data points in the training dataset and assigns the class label that is most common among its neighbors. The choice of K, distance metric (Euclidean, Manhattan, etc.), and weighting scheme (uniform or distance-based) are crucial factors that influence the model's performance. Afterward, its optimizer tries all combinations of the settings using a method called Grid Search Cross-Validation to determine

The best model is chosen based on its F1 score, as it is a measure of precision.

Euclidean Distance formula is:

$$d(x, y) = \sqrt{\sum_{i=1}^n (y_i - x_i)^2} \quad (1)$$

Manhattan Distance formula is:

$$d(x, y) = |x_1 - x_2| + |y_1 - y_2| \quad (2)$$

Minkowski Distance formula is:

$$d(x, y) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}} \quad (3)$$

2. Naive Bayes

Naive Bayes applies Bayes' theorem with the 'naive' assumption of the features' independence. During the training, mean and standard deviation of each feature for the best-performing ones. Each category are calculated by Naive Bayes principle which is an assumption of the feature independence. This is accomplished during prediction through computation of probability for the data point to belong to each class using Bayes' theorem and then finally selecting the class with the highest probability among them. Naive Bayes is valued for simplicity, scalability, and effectiveness in handling high dimensional data for classification tasks.

The formula is:

$$P(Y|X) = \frac{P(X|Y) \times P(Y)}{P(X)} \quad (4)$$

Where:

$P(Y|X)$ is posterior probability = Probability of hypothesis A on the observed event B.

$P(X|Y)$ is likelihood probability = Probability of the evidence given that the probability of a hypothesis is true.

$P(Y)$ is prior probability = Probability of hypothesis before observing the evidence.

$P(B)$ is marginal probability = Probability of evidence.

3. Support Vector Machine

SVM operates under the supervised learning paradigm where the points are mapped into the dimensions formed by the existing features. It is focused to classify information that involves locating a hyperplane that clearly distinguishes between classes that have been chosen in a feature space. This hyperplane is the classification boundary where it separates all the data points of different classes that have the largest margin between them. Through the process of tuning the position and direction of the hyperplane, SVM tries to achieve the highest accuracy in the classification. It is a robust instrument, that can be applied both for the linear and nonlinear classification tasks.

4. Decision Tree

Decision Tree classifier for predictive classification modeling. Decision tree is a model that uses tree like structure and sequentially takes decision on the basis of input features for a new data point and gives the outcome or class of that data point [16]. We first define the DecisionTreeClassifier with a fixed random state for reproducibility. Then we create a parameter grid containing potential values for hyperparameters like maximum depth, minimum samples per leaf, and splitting criterion. Using GridSearchCV, we perform a search over the parameter grid to find the optimal combination of hyperparameters that maximizes accuracy.

5. Random Forest

The Random Forest algorithm, which builds up several trees, randomly chooses some, and then combined their predictions together. It is a machine learning algorithm that aims to get better accuracy. Different trees will observe only a small portion of the space of data features so this boosts diversity. In a prediction, each tree casts a vote for its most certain class, and the final decision is made in accordance with the majority. The algorithm controls overfitting and serves well with substantially large data. GridSearchCV is applied to the hyperparameters optimization like the number of trees and features to be considered which further increases the accuracy of the model.

6. Logistic Regression

Logistic Regression is arguably one of the most popular machine learning algorithms applied to kinds of things where there is a binary outcome, like spam email classification. This technique applies a logistic curve that gives a probability of classifying a particular 'input'. Logistic regression is a statistical method which describes the relationship between a categorical dependent variable and one or more independent variables [17]. This implies that the algorithm "learns" the coefficient of features in the data and then predicts the value based on these coefficients on the new data.

It calculates the probability $P(Y = 1 | X)$ of the binary outcome Y (e.g. success/failure) given predictor variables X Using the logistic function (this is also known as sigmoid function)

The formula is:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n)}} \quad (5)$$

Where:

$P(Y = 1 | X)$ = is the probability of the outcome being success (1) given the predictor variable X .

e = it is the base of the natural logarithm.

$\beta_0, \beta_1, \dots, \beta_n$ = this are the coefficient if the logistic model

$X_1, X_2, \dots, X_n$ = those are the predictor variable.

$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n$ = this term is the linear combination of predictor variable and their corresponding coefficients.

7. Ensemble Model

- a) TabNet: It is a deep learning architecture specially developed for tabular data. It is a multi-layered process that includes feature selection and decision-making layers. The main innovation is the dynamic selection of informative features by means of attention mechanisms at each layer. TabNet is effective in this matter since it is able to concentrate on the vital features and forget about the irrelevant ones which helps it in accurately establishing the complex patterns in the tabular data sets.
- b) XGBoost: It is an ensemble learning algorithm that trains a set of different decision trees sequentially. XGBoost works on the basis of many fundamental decision trees, or weak learners, and calculates the quantity of similarity scores separately [18]. Every tree is taught to fix the mistakes by the previous trees, and hence end up with more accurate prediction. XGBoost employs a gradient descent optimization algorithm to minimize the objective function which is composed of a loss function that measures the difference between the predicted and actual values and regularization terms that penalize complex models to avoid their overfitting.
- c) Multilayer Perceptron: MLP is a kind of artificial neural network made up of multiple layers of neurons. The perceptron has a simple activation function, the sign of the sum of the weights and inputs, z , but it is able to perform only a limited amount of processing [19]. Each neuron performs a weighted sum of its inputs, adds a bias term, and sends the result through a nonlinear activation function. The output of one layer is the input to the next layer which is the way MLPs can learn complex nonlinear relations in the data [20]. The training of an MLP entails the adjustment of weights and biases via backpropagation by propagating errors backwards through the network to update the network parameters.

The equation is:

$$y = f\left(\sum_{i=1}^n w_i * x_i + b\right) \quad (6)$$

Where:

w_i = are the weights,

x_i = are the inputs,

b = is the bias and

f = is the activation function.

Ensemble Algorithm Steps:

- Firstly, initialize the TabNet, XGBoost and MLP models.
- Train each base model on the training data independently. In order to minimize the selected loss function, each parameter must be optimized.
- Use the trained base model to make predictions on the validation or test data set.
- Combine the predictions from TabNet, XGBoost, and MLP using a predefined method and averaging the predictions or weighted averaging based on performance.
- The output of the final ensemble model is the average of all the predictions by the individual models.
- Evaluate the effectiveness of the ensemble model by using accuracy, F1 score, precision, and recall as metrics.

IV. EVALUATION

In the last step of the prediction model, we have a detailed assessment of our model performance so that we use different evaluation metrics to better estimate our model. The metrics here include classification accuracy, confusion matrix analysis and the computation of F1-scores, among others. From this analysis we obtain important information on the performance and stability of the model in predicting outcomes. Several evaluation metrics are used to evaluate the considered feature selection and ranking methods as well as the considered classifiers [21]. Such a complete assessment helps to ensure that the outcomes of our model conform to reality and work as expected for practical purposes.

A. Confusion Matrix

In the realm of data analysis a confusion matrix serves as a tool to assess how well a classification model is performing. Figure 2 presents a summary of the model's predictions versus the actual class labels.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative(0)	FN	TN

Fig. 2. Confusion Matrix

Where:

TP (True Positive) is the number of instances that are positive and are classified correctly as positive.

FP (False Positive) is the number of instances that are actually negative but are incorrectly classified as positive.

TN (True Negative) is the number of instances that are negative and are classified correctly as negative.

FN (False Negative) is the number of instances that are actually positive but are incorrectly classified as negative.

B. Classification Accuracy

It checks the ratio of correctly classified instances among all the instances in the dataset.

The formula for classification accuracy is:

$$Accuracy = \frac{Number\ of\ Correct\ Prediction}{Total\ Number\ of\ Prediction} \quad (7)$$

In mathematical terms:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

C. Precision

It shows how accurately the model is able to find positive cases. In general, it shows us the ratio of the model's correct predictions among all the predicted positive classes.

The formula for precision is:

$$Precision = \frac{TP}{TP + FP} \quad (9)$$

D. Recall

The recall, also called sensitivity, is about the model's capability to correctly detect positive patterns. It determines the percentage of true positive instances correctly predicted by the model to all instances, which were in fact positive ones.

The formula for recall is:

$$Recall = \frac{TP}{TP + FN} \quad (10)$$

E. F1 Score

F1 score is being the harmonic mean of precision and recall. The purpose of F1 Score is to achieve a balance between recall and precision hence it serves as a worthy performance indicator, particularly for cases where the classes are not equally represented. The range for F1 Score is [0, 1].

The mathematical formula of F1 score is:

$$F1\ Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (11)$$

V. RESULT ANALYSIS

In this study, six machine learning and three deep learning algorithms were applied to predict diabetes. In deep learning combine those three deep learning models to make an ensemble model. The machine learning algorithms which we are using are KNN, NB, SVM, DT, RF, and LR. And deep learning algorithms are TabNet, XGB, and MLP. All these algorithms were applied in two different dataset PIMA Indian dataset and another dataset. In the classification and multi-classification study, different machine learning algorithms including SVM, NB, and RF have been used for diabetes prediction [22][23]. Data was divided into two portions, training data and testing data, both these portions consisting 80% and 20% data respectively. The primary evaluation criterion that we worked in this work was prediction accuracy. The algorithm's overall success rate is its accuracy.

According to the Table 2 the metrics such as F1 score, precision, recall, and accuracy, we performed a thorough study of six machine learning models that were employed to

predict whether the patient has diabetes or not. K-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest models demonstrated high accuracy on mentioned metrics, the F1 score ranging from 0.78 to 0.84 and the precision from 0.83 to 0.85 and recall values at 0.74 to 0.83 respectively.

Table 2. Model Average Performance Score for Dataset 1

Model Name	F1 Score	Precision	Recall	Accuracy
K Nearest Neighbor	0.804	0.854	0.797	87%
Naive Bayes	0.688	0.820	0.592	81%
Support Vector Machine	0.841	0.849	0.833	89%
Decision Tree	0.807	0.767	0.852	86%
Random Forest	0.784	0.833	0.741	86%
Logistic Regression	0.711	0.888	0.592	83%
TabNet + XGB + MLP	0.843	0.895	0.796	90%

Table 3. Model Average Performance Score for Dataset 2

Model Name	F1 Score	Precision	Recall	Accuracy
K Nearest Neighbor	0.756	0.803	0.715	84%
Naive Bayes	0.598	0.649	0.555	74%
Support Vector Machine	0.814	0.888	0.752	88%
Decision Tree	0.774	0.844	0.715	86%
Random Forest	0.810	0.860	0.766	88%
Logistic Regression	0.632	0.689	0.584	77%
TabNet + XGB + MLP	0.938	0.934	0.942	96%

These models scored even higher in terms of their accuracies that ranged from 86% to 89%. The hybrid deep learning ensemble model achieved high performance with the highest accuracy of 90% in dataset 1.

In Table 3, K Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest indicated a good performance of 84% to 88% accuracies in dataset 2. They achieved F1 scores ranging from 0.75 to 0.81 with precision from 0.80 to 0.88 and recall of 0.71 to 0.76, respectively. However, the hybrid deep learning ensemble model was able again to demonstrate high performance with the highest accuracy of 96%, followed by Random Forest model with an

accuracy of 88%. Nevertheless, figure 3 Naïve Bayes model exhibited a slightly lower accuracy rate of 81% in contrast to the performance on dataset 1. In the figure 4 the Naïve Bayes model demonstrated the lowest accuracy of 74% for dataset 2.

However, that NB, DT and LR had slightly below expected performance metrics than SVM for both of the dataset. In dataset 1 figure 3 the F1 scores ranging from 0.68 to 0.80, precision values around 0.76 to 0.88 and recall values around 0.59 to 0.85. And the accuracy range is 81% to 86%. And also in dataset 2, F1 scores ranging around 0.59 to 0.77, precision values around 0.64 to 0.84 and recall values around 0.55 to 0.71. The variations in the performance of NB, DT, and LR did not significantly affect their accuracy rates which were found to be satisfactory between 74% and 86%.

Moreover, the ensemble model built on TabNet, XGBoost (XGB), and Multi-Layer Perceptron (MLP) served as a promising option and produced a F1 score of 0.843, precision of 0.895, recall of 0.796 and accuracy of 90% for dataset 1 which is little less than dataset 2. And for dataset 2 figure 4, this ensemble model produced F1 score of 0.938, precision of 0.934, and recall of 0.94 and accuracy of 96%.

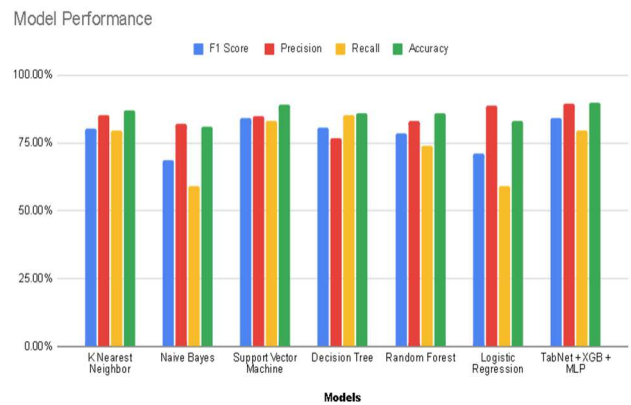


Fig. 3. Comparison Models for Dataset 1

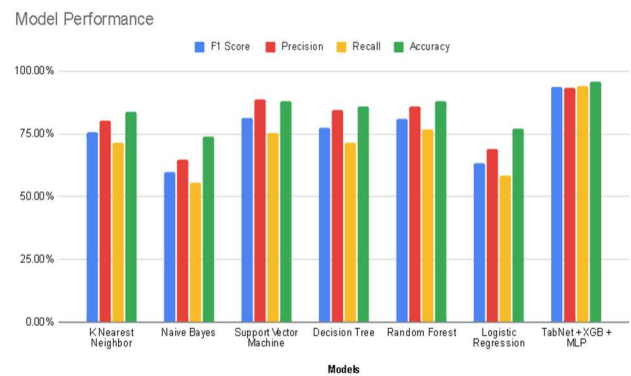


Fig. 4. Comparison Models for Dataset 2

Comparing all the model, the hybrid deep learning ensemble model surpassed other models in terms of accuracy for both datasets. However, different performance of Naive Bayes was observed between two datasets, and this may be caused by features of the data sets. Moreover, the Decision Tree model was stable in both datasets, thus it has shown the model can work well with any data distribution.

VI. CONCLUSION

The aim of our work was to examine many machine learning and deep learning algorithms that may be used for the prediction of diabetes diagnosis by employing both datasets separately. In our study, we evaluated six machine learning algorithms: These are k-nearest neighbor, decision tree, random forest, logistic regression, support vector machine and Naïve Bayes. Also, we used an ensemble technique incorporating MLP, TabNet, and XGBoost (XGB). It was seen that the proposed hybrid deep learning ensemble model has a higher accuracy as compare to the individual machine learning model with an accuracy of 90 to 96% in two datasets. In contrast, the machine learning models achieved accuracies ranging from 74% to 89% in two datasets. Our findings indicate that the method of combining deep learning models through an ensemble could help to enhance predictive accuracy in this case. For the purpose of achieving better model fitting and prediction accuracy, the next stage for us is to further optimize the feature extraction process with the help of autonomous deep learning techniques. The hybrid deep learning ensemble algorithm gives us better accuracy. The introduction of novel hybrid ensemble techniques represents a significant advancement in predictive modeling for diabetes. Beyond diabetes, these methods handle large-scale medical data efficiently and improve results in healthcare. Moreover, we plan to make a detailed study of ensembles that better suit our data with the purpose of achieving better performance and resilience.

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